MGU-UGP BSc (HONS) MICROBIOLOGY

IVSEMESTER

DSE- MG4DSEMBG201– Computational Biology and Bioinformatics

(2024 ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Time: 2 Hours **Max. marks: 70**

Part A
Very Short Answer Questions
Answer all questions
Each question carries 1 mark

1. Define primary nucleotide sequence database. [K][1]
2. What is SCOP? [K][1]
3. Explain Web Mining.[U][2]
4. Name one application of bioinformatics in drug discovery. [A][4]
5. How is Python executed from the command line? [U][3] (1 x5=5)

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 2 marks.

6. What is k-means clustering? [U][2]
7. What is a Python version?[K][3]
8. What is protein sequencing? [K][1]
9. What is target validation? [U][4]
10. How is Python executed from the command line?[U][3]
11. Write down the amino acid residues.[K][1]
12. List the applications of bioinformatics in drug discovery.[A][4]
13. Explain string concatenation. [U][3]
14. Point out the relationship between protein sequence and function?[U][1]
15. Define a naming convention in Python. [K][3]
16. Write down the expansion APRIORI. [K][2]
17. What is ligand-based drug design? [U][4] (2 x10=20)

Part C
Short Essay Questions
Answer any 5 questions
Each question carries 6 marks

18. Write down the features, advantages, and limitations of Python programming.[U][3]
19. Explain classification and clustering techniques with suitable examples. [A][2]

20. Explain the levels of protein structure and function. [U][1]
21. Describe molecular docking [A][4]
22. Write a short note on Web Mining, explaining Web Content. [U][2]
23. Explain the role in structure-based drug design and scoring functions.[A][4]
24. Mention one advantage of Python. [K][3]

(6x5=30)

Part D

Long Essay Questions

Answer any 1 question

Each question carries 15 marks

25. Discuss the fundamentals of Computational Biology and Bioinformatics.[U][1]
26. Describe computational approaches in drug design, including target identification. [U][4]

(15x1=15)